Towards Automated Cardiac Catheterization Laboratory Activation using Medical Text Classification

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***Abstract*— Ischemic heart disease is the leading cause of death and in acute presentation there is a need to reduce the time from getting to the hospital to the time of treatment. We worked on an automated system to identify patients that need cardiac catheterization in the mimic-iv-ext cardiac disease using various text classification methods. As a baseline, we used TF-IDF with logistic regression and compared it to Bio_ClinicalBERT [1], which catastrophically failed (0% recall) due to class imbalance. We improved on the BERT model by using a knowledge graph from TF-IDF features and ICD-10, which is essential for minority class medical prediction (23% prevalence). Our novel multi-agent system solved the token limit truncation problem and achieved 98.7% recall on MI patients, missing only 1 of 76 critical cases requiring urgent catheterization. This approach significantly outperformed knowledge graph-enhanced models (73.7%, McNemar's χ^2=90.48, p<0.001).**

Keywords: *ischemic heart disease, acute myocardial infarction, cardiac catheterization laboratory activation, multi-agent systems, BERT, TF-IDF, knowledge graph, classification, classification, class imbalance*

I. Introduction

Ischaemic Heart Disease is the leading cause of death world-wide responsible for 13% of all deaths [2]. Ischaemic Heart Disease presents in emergency cases as acute coronary syndromes (ACS) which has two main types that are ST elevation myocardial infarction (STEMI) and non ST elevation myocardial infarction (NSTEMI). In ACS, there is a need to reduce the time from the patient getting through the hospital door to the time of definitive treatment with balloon catheterization (door to balloon time) [3]. Artificial Intelligence systems have been implemented for automatic detection of STEMI from ECG only [4]. For example, the Queen of Hearts model was shown to be able to identify specific versions of STEMI better than physicians [5]. Linking these platforms to notification systems has been demonstrated to reduce the door to balloon time [4].

However, to decide to send the patient for emergency cardiac catheterisation it is important to consider the patient's presenting history, ECG results and the results of lab investigations. Our research looks at using a LLM to classify text in the Electronic Health Record (EHR) which includes the History of Presenting Illness (HPI), Physical Examination, ECG report and lab results (Troponin-T) to automatically make the decision and send a notification to the catheterisation lab. Essentially this is a binary medical text classification problem. There have been limited work focusing on the classification of patients that need cardiac catheterisation using text information from the EHR record. We aimed to combine text-based information into a single input for a transformer-based model.

Therefore, our research problems are:

- Can transformer-based models predict which patients need cardiac catheterization using text-based classification methods using information in the patient Electronic Health Record (HPI, Physical Examination, Chief Complaints, ECG report, lab results) for patients with possible ACS?
- Do knowledge graphs and Agent Systems improve performance of transformer classification for cardiac catheterization need for patients with possible acute coronary syndromes?

Our contributions are: (1) comprehensive study combining clinical narratives for catheterization treatment prediction in ACS patients, (2) investigating how domain knowledge integration is essential for minority class medical prediction; (3) multi-agent approach to solve the token truncation problem on BERT model.

II. Related Work

Traditional machine learning approaches for cardiac catheterization prediction have focused primarily on structured clinical variables. For example, Kutrani and Eltalhi [6] compared J48 Decision tree, k-Nearest Neighbors (kNN), Naïve Bayes (NB), and Support Vector Machine and noted that J48 had the best performance. Work has been done on classification of medical text using various Machine Learning models including gradient boosting with Text Frequency – Inverse Document Frequency (TF-IDF) and Transformers, which were comparable to gradient boosting [7]. Other studies have used Convolutional Neural Networks and Recurrent

Neural Networks for text classification, which have outperformed BERT for this task [8]. Knowledge graphs have been used to improve the performance of BERT like models on domain specific areas [9].

However, prior work has not addressed catheterization prediction using comprehensive EHR text (HPI, physical exam, ECG reports, labs). Most AI systems for acute coronary syndromes focus solely on ECG interpretation, ignoring clinical context essential for catheterization decisions. Our work fills this critical gap.

Recently LLMs have been shown to perform well in medical text classification [10]. LLM performance improves as the model size increases [11]. Using a model pretrained with similar context for the application will lead to better performance. Hence, in this study we selected Bio_ClinicalBERT [1] over BioBERT and PubMedBERT because its pretraining on MIMIC-III clinical notes better matches our target domain.

While BERT achieves powerful contextual embeddings through unsupervised pretraining, it lacks explicit domain knowledge [12], particularly problematic for minority class medical events where pattern frequency is low. To improve on this shortcoming, we use a knowledge graph. Integrating knowledge graphs with BERT has been shown to provide explicit biomedical knowledge [13], enhance generalization and support interpretability [9]. BERT has a limited context window as a way to ensure that all the information in the clinical notes was utilized for the classification. Studies have shown that LLM can be used in clinical text summarization with human level accuracy [14].

III. Method

We utilized the MIMIC-IV-EXT cardiac disease dataset (version 1.0.0) [15, 16] from MIT's Laboratory for Computational Physiology, comprising 4,761 patients with cardiovascular conditions. From this cohort, we extracted clinical narratives including chief complaints, history of present illness (HPI), physical examination notes, ECG reports (clinician-generated), and laboratory results, especially Troponin. We convert the troponin numerical results into 4 categories: elevated/normal/not detected/not tested. Troponin test is the most sensitive test to indicate myocyte injury as happens in myocardial infarction [17].

To prevent data leakage since the HPI was taken from discharge summary, we implemented pattern-based detection to identify inadvertent mentions of cardiac catheterization procedures in clinical notes (e.g. "cath lab" or "angiography"). Patients with such mentions were excluded to prevent the model from learning post-procedural information rather than making predictions from pre-procedural data. As a result, 1,903 patients with data leakage were excluded (1,274 with catheterization, 629 without), leaving 2,858 patients for further filtering. After applying chest pain filtering criteria, 2,381 patients remained for final analysis.

Thus, patients were included if they had: (1) complete documentation (HPI, chief complaint, physical examination, ECG), (2) symptoms consistent with acute coronary syndrome (chest pain, angina, or substernal discomfort identified via regular expression pattern matching), and (3) no evidence of data leakage. This yielded 2,381 patients (30.4% catheterization prevalence), partitioned into training (n=1,904, 80%) and test (n=477, 20%) sets via stratified sampling (random seed: 42). The training set contained 581 catheterization cases (30.5%) and 402 MI patients (21.1%); the test set contained 145 catheterization cases (30.4%) and 112 MI patients (23.5%).

After preprocessing, we developed and compared four approaches: (1) Term Frequency - Inverse Document Frequency (TF-IDF) with Logistic regression as a baseline; (2) Bio_ClinicalBERT (vanilla model) with fine-tuning; (3) Bio_ClinicalBERT + Knowledge Graph (KG); (4) Multi-Agent System (LLM + KG + model from approach 3).

Our first approaches utilized TF-IDF vectorization with max 5,000 features, unigram, and bigrams. It was then followed by balanced logistic regression served as the traditional machine learning baseline.

The second approach was to fine-tune the Bio_ClinicalBERT model on concatenated clinical text (HPI + physical exam + ECG + troponin status). The training parameters were as follows: 4 epochs, learning rate 2e-5, batch size 16, class-weighted cross entropy loss, OneCycleLR scheduler, dropout 0.1, and early stopping (patience=3) based on validation F1 score. The model worked below what the baseline achieved, so we were thinking about another solution to increase the model's performance.

A knowledge graph incorporated 20+ patterns across five categories: ECG findings (10 patterns), troponin status (3 categories), symptoms (chest pain patterns), medication (aspirin/nitroglycerin), and alternative diagnoses (atrial fibrillation, pericarditis, etc.) that were retrieved from TF-IDF features. The top 7 patterns by probability were included per patient to manage input length. The way we formed the knowledge graph construction was: for each pattern p (e.g., 'st_elevation'), we utilized Laplace smoothing: $P(CATH|p) = (n\_cath + 1) / (n\_total + 2)$. Confidence scores were computed as $\min(n\_total/50, 1.0)$ to down weight rare patterns. In this stage, we found that the context window became our limitation as the information about the patient and the knowledge graph have taken more than the token limit for the BERT's input. That led us to develop another approach.

The last approach was to use multi-agents for information retrieval in order to weigh and prioritize the information given to LLM. A three-agent architecture consisted of three agents. Agent 1 (Clinical Reasoner) supported by Microsoft Phi-3-mini-4k-instruct [18] analyzed patient data using an ICD-10 hierarchical knowledge graph (codes I20-I25 for ischemic heart disease [19]). The LLM extracted key findings, assigned ICD-10 codes, assessed MI likelihood, and generated feature importance weights (JSON output, temperature=0.1, greedy decoding). Agent 2 (Input Constructor) prioritized and truncated input fields proportional to Agent 1's feature importance weights to fit the 512-token limit, ensuring high-priority features (e.g., ECG in MI cases) were preserved. Given the weights, we allocated tokens proportionally in a sense of tokens_i = 450 x weight_i, truncating each field to its allocation to fit the token limit for BERT. Agent 3 (Final Classifier) utilized pre-trained KG checkpoint (from approach 3) and adjusted its output probabilities based on Agent 1's clinical reasoning (+0.15 for MI diagnosis, +0.10 for urgent ICD-10 codes [I21/I22/I24], +0.05 per high-risk finding). The probability adjustments allow the system to systematically incorporate explicit clinical domain knowledge (e.g., MI urgently requires catheterization) identified by the LLM into ClinicalBERT's predictions, creating a hybrid approach that combines neural pattern recognition with interpretable rule-based clinical reasoning rather than relying on the neural network's opaque learned associations alone.

Lastly, all models were evaluated on the held-out test set using ROC-AUC, accuracy, precision, and recall. Critically, we performed MI-stratified evaluation to assess performance on myocardial infarction patients (ICD-10 codes I21/I22), for whom catheterization decisions are time-sensitive. We reported CATH recall (sensitivity), F-1 score for CATH class, and ROC-AUC for this high-risk subgroup. All experiments used identical train/validation/test splits for fair comparison. Our north-star metric is the CATH recall, yet additional metrics were provided for the sake of completeness. Missing patients who need catheterization (false negatives) can lead to myocardial infarction complications or death, whereas unnecessary catheterization (false positives) is an invasive but safe diagnostic procedure. This procedure sometimes was done as a diagnostic tool for the clinicians [20]. Therefore, we prioritized recall over precision. Ultimately, all models were evaluated with 2,000-iteration bootstrap confidence intervals (95% CI) and compared using McNemar's test for paired classifiers.

IV. Result

Five classification approaches were evaluated on the test set. The multi-agent system achieved significantly higher recall (88.3% [82.7-93.2%]) than the ClinicalBERT with KG (56.6% [48.2-64.6%], McNemar's χ^2=90.48, p<0.001), TF-IDF (66.9% [59.2-74.3%], χ^2=94.01, p<0.001), and vanilla ClinicalBERT (0%, χ^2=128.00, p<0.001). Among 112 MI patients requiring urgent catheterization decisions, the multi-agent system achieved superior sensitivity: 98.7% CATH recall (75 of 76 detected, only 1 missed case) compared to ClinicalBERT with KG (73.7%, 20 missed), TF-IDF (77.6%, 17 missed), and vanilla ClinicalBERT (0%, 76 missed). This improvement came at the cost of reduced precision (34.0% vs 62.1% for Enhanced KG), reflecting an intentional sensitivity-optimized strategy. While ClinicalBERT with KG optimized balanced performance and obtained a better accuracy, multi-agent prioritized sensitivity for time-critical cases and focused on better CATH recall.

The TF-IDF baseline (the first approach) achieved the highest overall ROC-AUC (0.806), with 75.7% accuracy and 66.9% CATH recall. Top predictive features for catheterization included "heparin," "substernal chest," and "stress". On the other hand, "atrial fibrillation" and "flutter" were negatively associated with catheterization decisions because it led to another diagnosis.

Our second approach, Bio_ClinicalBERT without knowledge graph, failed catastrophically (ROC-AUC 0.581, 69.6% accuracy), predicting no catheterization for all patients (0% CATH recall). This demonstrates the model's inability to learn from severe class imbalance (23.3% CATH prevalence). Without additional knowledge structures, the model performed very badly.

Knowledge graph integration substantially improved performance. The approach further improved to 75.1% ROC-AUC, 76.3% accuracy, and 56.6% CATH recall. The performance matched TF-IDF baseline performance while providing interpretable clinical reasoning.

TABLE I
EVALUATION METRICS OF OUR APPROACHES

Approaches	All Patients			Myocardial Infarction patients		
	Recall (CATH)+	F-1 score	ROC-AUC	Recall (CATH)+	F-1 score	ROC-AUC
TF-IDF + Logistic Regression	0.67 [0.59-0.74]	**0.63**	**0.81 [0.76-0.85]**	0.78	0.73	0.70
Bio_ClinicalBERT (vanilla)	0.00	0.00	0.58 [0.52-0.64]	0.00	0.00	0.58
Bio_ClinicalBERT + Knowledge Graph	0.57 [0.48-0.65]	0.59	0.75 [0.70-0.80]	0.74	**0.80**	**0.67**
Multi Agent (LLM + KG + ClinicalBERT)	**0.88* [0.83-0.93]**	0.49	0.73 [0.68-0.78]	**0.98**	**0.80**	**0.67**

Values shown as point estimate [95% confidence interval] from 2,000 bootstrap iterations.
**) statistically significant at p<0.001*
+) north-star metrics

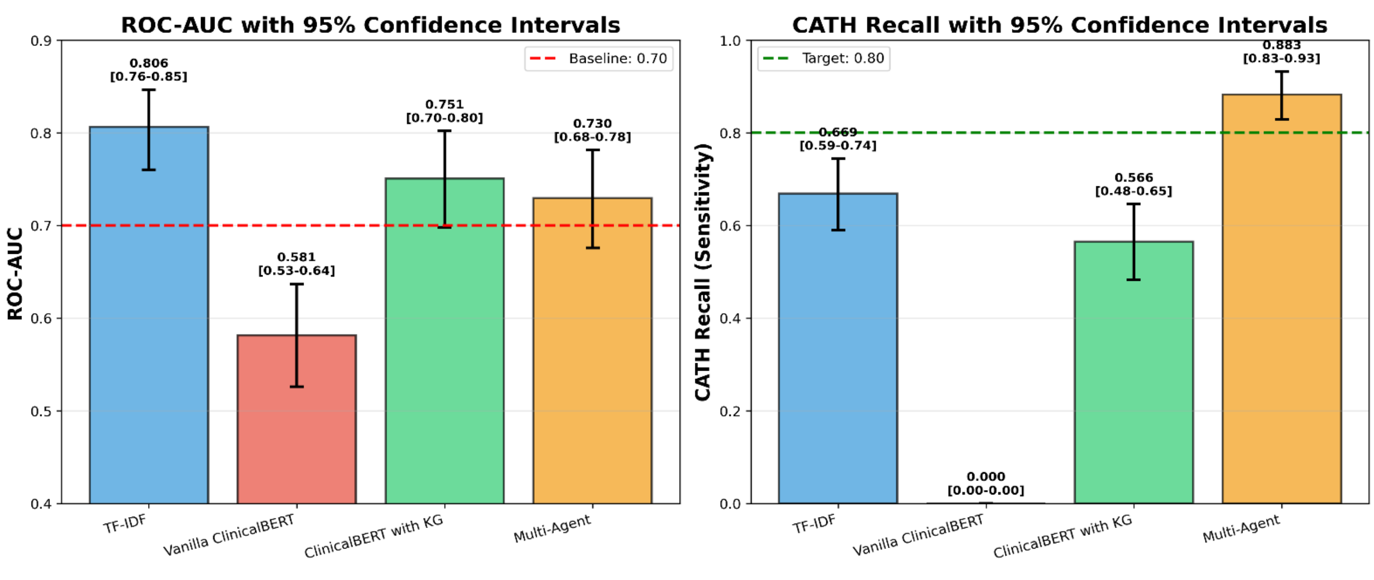

Fig. 1. Model Comparison on ROC-AUC and CATH Recall

V. DISCUSSION

Our multi-agent system's 98.7% MI recall (1 missed case vs 20 for ClinicalBERT with KG) represents a clinically meaningful improvement for time-critical decisions. In emergency cardiology, the door to balloon golden time catheterization is crucial for minimizing myocardial damage. While the system's reduced precision (34.0% vs 62.1%) results in more unnecessary catheterizations.

ClinicalBERT with KG and Multi-Agent showed similar AUC (McNemar's test p<0.001 despite similar discrimination) indicates they represent different operating points on the ROC curve rather than different discrimination abilities. The multi-agent system deliberately shifts the decision threshold to prioritize sensitivity, which is appropriate for the clinical context where false negatives have far greater consequences than false positives.

The catastrophic failure of vanilla ClinicalBERT (0% recall) demonstrates that the transformer model cannot handle severe class imbalance (30% prevalence) without domain knowledge integration. This finding contradicts assumptions that pretrained language models can adapt to any domain through fine-tuning alone, highlighting the necessity of explicit knowledge structures for minority class medical events.

Bootstrap confidence intervals (n=2,000 iterations) showed substantial uncertainty in recall estimates, particularly for multi-agent (88.3% [82.7-93.2%]), indicating the need for validation on larger test sets. The single-center, retrospective nature limits generalizability, and external validation is required before clinical deployment. Additionally, the 66% increase in false positives must be validated against cost-effectiveness and resource utilization in real-world clinical settings.

VI. Conclusion

Among the four approaches we have done, the multi-agent system for predicting cardiac catheterization needs in ACS patients achieved 98.7% recall on MI cases. This approach missed only 1 of 76 patients requiring urgent intervention, compared to 20 missed by the knowledge graph baseline (McNemar's $p<0.001$). This represents a dramatic improvement over vanilla transformers (0% recall) and demonstrates that incorporating structured clinical knowledge through LLM-guided reasoning is essential for minority class medical predictions with severe class imbalance.

Vanilla Bio_ClinicalBERT failed completely on imbalanced medical data highlighting the need for domain knowledge integration. Knowledge graph integration improved recall to 56.6%, approaching TF-IDF performance (66.9%) while providing interpretable clinical reasoning. The multi-agent architecture solved the token context window limitation through LLM-guided feature prioritization, that translates to 19 fewer missed life-threatening events in our test cohort.

The future work should: (1) validate on external datasets, (2) extend to real-time ECG integration for multimodal implementations, and (3) include prospective trials with clinician feedback on the acceptable false positive rate for emergency MI detection.

Reproducibility

All code that has been used for this paper is available here: https://github.com/ignsagita/cath_pred.

The dataset can be accessed with permission here: https://physionet.org/content/mimic-iv-ext-cardiac-disease/1.0.0/.

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